Risk-Sensitive Markov Decision Processes

Finite-Horizon Framework for Risk-Averse Dynamic Programming

Based on: Ruszczyński (2010) & Ruszczyński–Shapiro (2006)

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1 Introduction

In classical stochastic control, finite-horizon Markov Decision Processes (MDPs) are built around optimising an expected value. The standard risk-neutral objective seeks a policy Π that minimises the expected cumulative cost:

$$\min_{\Pi} \mathbb{E}\left[\sum_{t=1}^{T} c_t(x_t, u_t)\right]. \tag{1}$$

The expectation operator is the sole mechanism mapping uncertain future costs back to the perspective of each current time step. Relying on expectation alone, however, masks the variance and the threat of rare but extreme outcomes.

Consider two cost distributions: one with a 50% chance of costing \$0 and a 50% chance of costing \$100, and another with a guaranteed cost of \$50. An expectation-based model treats these as identical. In high-stakes environments such as financial portfolio management, an extreme downside is unacceptable even at equal expected cost. Decision-makers in these domains are risk-averse: willing to pay a premium against worst-case scenarios even if this raises the overall expected cost. Moving from a risk-neutral to a risk-sensitive MDP formulation is essential for generating policies that account for downside risk.

Resolving this requires two steps. The first is establishing what a well-behaved one-step risk operator looks like. The second is showing how such operators can be composed into a full dynamic programming recursion without losing the Markov structure. Ruszczyński and Shapiro [2006] address the first step by introducing an axiomatic definition of conditional risk mappings: convex, monotone operators that extend the conditional expectation to a nonlinear setting. Their paper already constructs the recursive composition and derives dynamic programming equations from it. Ruszczyński [2010] then adapts this foundation to controlled Markov processes. The key contribution is the risk transition mapping σ, which restricts the adversary's ambiguity set to depend only on the current state and action. This restores the Markov property and makes backward induction tractable.

The primary results are:

1. **Axiomatic foundation [Ruszczyński and Shapiro, 2006]:** conditional risk mappings satisfying axioms (A1)–(A3) admit a duality representation as worst-case conditional expectations, and compose recursively to form multi-period risk objectives.

2. **Characterisation of time consistency (Theorem 1 of Ruszczyński 2010):** a dynamic risk measure is time-consistent if and only if it admits a nested one-step factorisation. This is the risk-averse analogue of the tower property and restores Bellman backward induction.

3. **Risk-averse Bellman equations (Theorem 2 of Ruszczyński 2010):** replacing the conditional expectation with the risk transition mapping σ yields a well-posed recursive optimality equation. Under mild regularity conditions, an optimal Markov policy always exists.

2 System Model

2.1 Description of the Model

The mechanics of the system analysed by Ruszczyński [2010] follow the standard discrete-time controlled Markov framework. For a finite horizon T, the system is defined by a state space $\mathbb{X}$ and a control space $\mathbb{U}$. At each time step t, the decision-maker observes the state $x_t \in \mathbb{X}$ and selects an action u_t from the admissible set $U_t(x_t)$. The system evolves under a controlled

transition kernel $Q_t(\cdot \mid x_t, u_t)$. Each state-action pair incurs a stage cost $Z_t = c_t(x_t, u_t)$ for $t = 1, \ldots, T$, plus a terminal cost $Z_{T+1} = c_{T+1}(x_{T+1})$.

Instead of minimising a cumulative expected value, the objective is to minimise a nested, time-consistent dynamic risk measure $J(\Pi)$. A dynamic risk measure $\{\rho_{t,T}\}_{t=1}^{T}$ is time-consistent if, for any two future cost sequences Z and W that are identical over $[\tau, \theta)$, the condition $\rho_{\theta,T}(Z_\theta, \ldots, Z_T) \leq \rho_{\theta,T}(W_\theta, \ldots, W_T)$ implies $\rho_{\tau,T}(Z) \leq \rho_{\tau,T}(W)$. This requirement drives the objective to the recursive form:

$$J(\Pi) = c_1(x_1, u_1) + \rho_1\big(c_2(x_2, u_2) + \rho_2\big(c_3 + \cdots + \rho_{T-1}(c_T + \rho_T(c_{T+1}))\big)\big), \tag{2}$$

where each ρ_t evaluates the risk of the future cost sequence from the perspective of time t.

2.2 Relation to Course Concepts and Static Risk Measures

As established in Mahajan [2026], the underlying dynamics of this model (states, actions, and controlled state transitions) are identical to the standard stochastic control problems covered in class. The key difference is the evaluation metric. In the risk-neutral setting, the operator mapping future costs backward is the conditional expectation $\mathbb{E}[\cdot \mid \mathcal{F}_t]$. In the framework of Ruszczyński and Shapiro [2006] and Ruszczyński [2010], this linear operator is replaced by a convex, monotone conditional risk operator ρ_t.

Static risk measures map a random cost Z to a single risk-adjusted number. Common examples include:

Value at Risk (VaR). For confidence level $\alpha \in (0, 1)$:

$$\mathrm{VaR}_\alpha(Z) = \inf\{z \in \mathbb{R} \mid \mathbb{P}(Z \leq z) \geq \alpha\}. \tag{3}$$

VaR is simple but not subadditive, and blind to loss severity beyond the threshold.

Conditional Value at Risk (CVaR). For continuous distributions:

$$\mathrm{CVaR}_\alpha(Z) = \mathbb{E}[Z \mid Z \geq \mathrm{VaR}_\alpha(Z)]\,. \tag{4}$$

CVaR averages the worst $(1-\alpha)$ fraction of outcomes and is subadditive. Its tractable definition via optimisation is given in Rockafellar and Uryasev [2000].

Entropic Risk Measure. With risk-sensitivity parameter $\theta > 0$:

$$\rho(Z) = \frac{1}{\theta} \log \mathbb{E}\left[e^{\theta Z}\right]. \tag{5}$$

Static risk measures break down in dynamic problems because they are not time consistent. Applying VaR or CVaR to the sum of all future costs creates a fixed plan based on the initial perspective. When new information arrives at $t = 1$, recalculating the same measure for remaining costs frequently yields different preference rankings. An agent could commit to an optimal policy at $t = 0$ and abandon it at $t = 1$ despite no unexpected events occurring, breaking Bellman's principle. As a concrete illustration: a VaR-optimal plan at $t = 0$ may rank two future scenarios in one order. After observing the first-stage outcome, recalculating VaR for the remaining horizon can reverse that ranking, causing the agent to abandon the original plan even when nothing unexpected occurred.

The Entropic Risk Measure is a notable exception. The exponential function turns addition of costs into multiplication of expectations, giving a clean one-step factorisation that does not depend on the full cost history. This unique structure is discussed in Section 31 of Mahajan [2026]. A general theory is needed for other risk measures, and that is what Ruszczyński and Shapiro [2006] and Ruszczyński [2010] provide.

2.3 Key Conceptual and Technical Difficulties

Incorporating risk into the DP framework requires resolving two challenges.

Time Consistency. Bellman's principle requires the global criterion to decompose recursively: an optimal plan from any intermediate state must coincide with the globally optimal plan. The first challenge is identifying a mathematical structure for risk measures that guarantees this decomposition.

History Independence. Even with a time-consistent structure, conditional risk measures are $\mathcal{F}_t$-measurable by definition, so risk evaluation at time t may depend on the entire process history $(x_1, \ldots, x_t)$. To see why this matters: the set $\mathcal{A}_t$ of adversarial measures is $\mathcal{F}_t$-measurable by construction. This means the worst-case distribution the adversary selects at time t can depend on the entire path $(x_1, \ldots, x_t)$, not just the current state. An agent defending against this adversary must remember the whole history, so no Markov policy can be optimal in general. If the optimal action u_t depends on the full trajectory, the state-based DP decomposition breaks down.

3 Main Results

To reach the risk-averse Bellman equations, we trace the logical progression across Ruszczyński and Shapiro [2006] and Ruszczyński [2010]. We first establish why the objective takes the nested form of one-step conditional mappings, then show how abstract risk is computed via a duality representation, and finally demonstrate how the Markov property is restored.

3.1 Constructive and Axiomatic Foundations

The two papers approach the structural requirement from complementary directions.

Ruszczyński and Shapiro [2006] take a constructive approach. They define the required properties for a one-step conditional risk mapping and construct the multi-period objective by composing these operators. Let $\mathcal{Z}_t$ denote the space of $\mathcal{F}_t$-measurable random variables.

Definition (Conditional Risk Mapping; Ruszczyński and Shapiro 2006). A functional $\rho_t : \mathcal{Z}_{t+1} \to \mathcal{Z}_t$ is a conditional risk mapping if it satisfies the following axioms for all $Z, W \in \mathcal{Z}_{t+1}$:

(A1) **Convexity:** $\rho_t(\alpha Z + (1-\alpha)W) \leq \alpha\rho_t(Z) + (1-\alpha)\rho_t(W)$ for all $\alpha \in [0,1]$.

(A2) **Monotonicity:** If $Z \leq W$ a.s., then $\rho_t(Z) \leq \rho_t(W)$.

(A3) **Translation Equivariance:** $\rho_t(Z + V) = \rho_t(Z) + V$ for all $V \in \mathcal{Z}_t$.

(A4) **Positive Homogeneity:** $\rho_t(\alpha Z) = \alpha\rho_t(Z)$ for all $\alpha \geq 0$.

Axioms (A1)–(A3) are the core of the 2006 paper. Positive homogeneity (A4) is an additional condition adopted by Ruszczyński [2010] for the MDP setting. Together (A1) and (A4) imply subadditivity, making the mapping coherent in the sense of Artzner et al. [1999].

Ruszczyński and Shapiro [2006] already define the nested composition in Section 7 as a constructive starting point and derive dynamic programming equations from it. Ruszczyński [2010] goes further and asks the converse question: what properties must a dynamic risk measure satisfy for this nested structure to be the only time-consistent form? Theorem 1 answers this, showing that conditions (7) and (8) together with time consistency are equivalent to the nested factorisation.

Definition (Time Consistency; Ruszczyński 2010). A dynamic risk measure $\{\rho_{t,T}\}_{t=1}^{T}$ is time-consistent if, for all $1 \le \tau \le \theta \le T$ and all cost sequences Z, W agreeing on $[\tau, \theta-1]$, the condition

$$\rho_{\theta,T}(Z_\theta, \ldots, Z_T) \le \rho_{\theta,T}(W_\theta, \ldots, W_T) \quad \text{a.s.} \tag{6}$$

implies $\rho_{\tau,T}(Z_\tau, \ldots, Z_T) \le \rho_{\tau,T}(W_\tau, \ldots, W_T)$. If two plans incur identical costs up to θ and one is preferred from θ onward, it must be preferred from today as well.

To link this generalised dynamic measure to the one-step mappings, Ruszczyński [2010] requires two regularity conditions.

Condition (7) — Translation Equivariance:

$$\rho_{t,T}(Z_t, Z_{t+1}, \ldots) = Z_t + \rho_{t,T}(0, Z_{t+1}, \ldots). \tag{7}$$

A cost known at time t shifts risk by exactly that amount, without interacting with future uncertainty.

Condition (8) — Normalisation:

$$\rho_{t,T}(0, 0, \ldots, 0) = 0. \tag{8}$$

Zero costs carry zero risk. Together with (7), the measure evaluates only future uncertainty.

Theorem (Nested Structure; Theorem 1 of Ruszczyński 2010). Under conditions (7)–(8), the dynamic risk measure $\{\rho_{t,T}\}$ is time-consistent if and only if for all $\tau \le \theta \le T$:

$$\rho_{\tau,T}(Z_\tau, \ldots, Z_T) = \rho_{\tau,\theta}\left(Z_\tau, \ldots, Z_{\theta-1}, \underbrace{\rho_{\theta,T}(Z_\theta, \ldots, Z_T)}_{\text{pre-collapsed tail}} \right). \tag{9}$$

Applying this repeatedly from T down to $\tau = 1$ gives the one-step factorisation:

$$\rho_{1,T}(Z_1, \ldots, Z_T) = Z_1 + \rho_1\big(Z_2 + \rho_2\big(\cdots + \rho_{T-1}(Z_T)\big)\big). \tag{10}$$

Proof sketch.
($\Rightarrow$) Suppose the measure is time-consistent. Construct Z and W agreeing on $[\tau, \theta-1]$, with $W_\theta = \rho_{\theta,T}(Z_\theta, \ldots, Z_T)$ and $W_k = 0$ for $k > \theta$. By conditions (7)–(8):

$$\rho_{\theta,T}(W_\theta, 0, \ldots) = \rho_{\theta,T}(Z_\theta, \ldots, Z_T).$$

Time consistency then forces $\rho_{\tau,T}(Z) = \rho_{\tau,T}(W)$, which is equation (9).
($\Leftarrow$) Suppose (9) holds. Take Z and W satisfying condition (4), so $\rho_{\theta,T}(Z_\theta, \ldots, Z_T) \le \rho_{\theta,T}(W_\theta, \ldots, W_T)$. Applying (9) to both sides converts this into a statement about $\rho_{\tau,\theta}$. Monotonicity of $\rho_{\tau,\theta}$ (axiom A2) then gives $\rho_{\tau,T}(Z) \le \rho_{\tau,T}(W)$, which is time consistency. $\square$

Significance. Equation (10) is the risk-averse analogue of the tower property of conditional expectation. The full-horizon risk measure collapses into a right-to-left chain of one-step evaluations. Each ρ_t compresses the tail into a single measurable number that feeds into ρ_{t-1}. This restores Bellman backward induction.

3.2 The Representation Theorem: Risk as Adversarial Optimisation

The nested structure guarantees time consistency, but leaves a practical question: how is an abstract conditional risk mapping ρ_t computed inside an algorithm? Ruszczyński and Shapiro

[2006] answer this via a duality representation theorem based on Fenchel-Moreau conjugate duality.

Theorem (Representation; Theorem 3.1 of Ruszczyński and Shapiro 2006). Let $\rho = \rho_{\mathcal{X}_2|\mathcal{X}_1}$ be a lower-semicontinuous conditional risk mapping satisfying (A1)–(A3). Then for all $\omega \in \Omega$ and $X \in \mathcal{X}_2$:

$$\rho_\omega(X) = \sup_{\mu \in \mathcal{P}_{\mathcal{Y}_2|\mathcal{F}_1}(\omega)} \big\{ \langle \mu, X \rangle - \rho^*(\mu, \omega) \big\}, \tag{11}$$

where $\rho^*(\mu, \omega)$ is the conjugate of ρ_ω and $\mathcal{P}_{\mathcal{Y}_2|\mathcal{F}_1}(\omega)$ is the set of conditional probability measures defined in Ruszczyński and Shapiro [2006]. If additionally (A4) holds, the penalty collapses to an indicator and the representation becomes:

$$\rho_\omega(X) = \sup_{\mu \in \mathcal{A}(\omega)} \langle \mu, X \rangle. \tag{12}$$

Proof sketch. The proof applies the Fenchel-Moreau theorem to ρ_ω. Since ρ_ω is convex and lower semicontinuous, it equals its biconjugate:

$$\rho_\omega(X) = \sup_{\mu} \big\{ \langle \mu, X \rangle - \rho^*_\omega(\mu) \big\}.$$

Axiom (A3) forces every μ in the effective domain of ρ^*_ω to be a conditional probability measure on $\mathcal{F}_2$ given $\mathcal{F}_1$, establishing that the supremum is taken over $\mathcal{P}_{\mathcal{Y}_2|\mathcal{F}_1}(\omega)$ as in equation (11). Adding positive homogeneity (A4) collapses ρ^* to an indicator function: zero inside the risk envelope $\mathcal{A}(\omega)$ and $+\infty$ outside. This gives the simpler form (12). □

The intuition is a minimax game. The decision-maker chooses an action resulting in uncertain cost X. Nature selects the measure μ that maximises expected cost, paying a penalty ρ^* for distorting the true distribution. The agent's risk is the resulting worst-case expected cost.

Proposition 4.1 and Theorem 4.1 of Ruszczyński and Shapiro [2006] further show that the adversarial measures in $\mathcal{A}(\omega)$ are valid conditional probabilities, giving:

$$\rho_t(X) = \sup_{\mu \in \mathcal{A}_t} \mathbb{E}_\mu[X \mid \mathcal{F}_t]. \tag{13}$$

In an MDP context, $\mathcal{A}_t$ is a set of transition kernels around the nominal Q_t. A risk-neutral agent has $\mathcal{A}_t = \{Q_t\}$, a singleton. A risk-averse agent uses a larger set, allowing Nature to shift transition probabilities adversarially. The size of $\mathcal{A}_t$ governs the degree of pessimism.

3.3 Rescuing the Markov Property: The Risk Transition Mapping

Directly applying ρ_t to an MDP introduces the history-dependence problem from Section 2. The set $\mathcal{A}_t$ is $\mathcal{F}_t$-measurable, so the adversary's choice of worst-case measure may depend on the full path $(x_1, \ldots, x_t)$. Without further restriction, $\mathcal{A}_t(\omega)$ can vary with the full history, and the adversary's choice is therefore not a function of the current state alone. Definition 5 of Ruszczyński [2010] breaks this dependence by requiring the risk envelope to be a function only of the current state x and the nominal kernel $Q(x, u)$, making the adversary's choice state-dependent but history-independent.

Definition (Risk Transition Mapping; Definition 5 of Ruszczyński 2010). A measurable functional $\sigma_t : \mathcal{V} \times \mathbb{X} \times \mathcal{M} \to \mathbb{R}$ is a risk transition mapping if:

$$\sigma_t(v, x, Q_t(x, u)) = \sup_{\mu \in \mathcal{A}(x,\, Q_t(x,u))} \langle v, \mu \rangle, \qquad \langle v, \mu \rangle = \int_{\mathbb{X}} v(y)\, \mu(dy), \tag{14}$$

where the risk envelope $\mathcal{A}(x, Q_t(x, u))$ depends only on the current state x and the nominal kernel $Q_t(x, u)$.

Definition (Markov Risk Measure; Definition 6 of Ruszczyński 2010). A one-step mapping ρ_t is a Markov risk measure if there exists a risk transition mapping σ_t such that

$$\rho_t(v(x_{t+1})) = \sigma_t(v,\, x_t,\, Q_t(x_t, u_t)).$$

By restricting the adversary's ambiguity set to depend only on (x_t, u_t), Definition 6 severs the dependence on history and restores the Markov property. It is the bridge between the abstract theory of Ruszczyński and Shapiro [2006] and a computable, state-dependent DP recursion.

3.4 The Risk-Averse Bellman Equations

With time consistency guaranteed by the nested structure and the Markov property restored by the risk transition mapping, Ruszczyński [2010] presents the main result.

Theorem (Finite Horizon; Theorem 2 of Ruszczyński 2010). Assume the following conditions hold for $t = 1, \ldots, T$:

(i) $Q_t(x, \cdot)$ is continuous for every $x \in \mathbb{X}$.

(ii) ρ_t is a Markov risk measure and $\mathcal{A}_t(x, \cdot)$ is lower semicontinuous for every $x \in \mathbb{X}$.

(iii) $x \mapsto c_t(x, u_t(x))$ and $c_{T+1}(\cdot)$ lie in $\mathcal{V}$ for all measurable selections u_t.

(iv) $c_t(x, \cdot)$ is lower semicontinuous for every $x \in \mathbb{X}$.

(v) $U_t(x)$ is compact for every $x \in \mathbb{X}$.

Define $v_{T+1}(x) = c_{T+1}(x)$ and for $t = T, \ldots, 1$:

$$v_{T+1}(x) = c_{T+1}(x), \tag{15}$$

$$v_t(x) = \min_{u \in U_t(x)} \big\{ c_t(x, u) + \sigma_t(v_{t+1},\, x,\, Q_t(x, u)) \big\}. \tag{16}$$

Then v_t is the unique optimal value function, and an optimal Markov policy $\hat{\Pi} = \{\hat{\pi}_1, \ldots, \hat{\pi}_T\}$ exists with:

$$\hat{\pi}_t(x) \in \arg\min_{u \in U_t(x)} \big\{ c_t(x, u) + \sigma_t(v_{t+1},\, x,\, Q_t(x, u)) \big\}. \tag{17}$$

Proof sketch. The proof proceeds by backward induction. Monotonicity (A2) allows the minimisation over u_t to be moved inside the risk mapping, isolating the one-step subproblem at each t. Replacing the abstract ρ_t with the Markov risk measure σ_t ensures v_t depends only on x_t.

Conditions (iv)–(v) guarantee the argmin is attained via the Weierstrass theorem. Conditions (i)–(iii) ensure σ_t inherits sufficient regularity from $\mathcal{A}_t$: this is not automatic since

σ_t is a nonlinear supremum over a set of kernels, unlike the linear expectation it replaces. Condition (iii) is specifically what ensures that after each backward step the new value function v_t remains in the space $\mathcal{V}$. Since σ_t takes its first argument from $\mathcal{V}$, if v_{t+1} were outside $\mathcal{V}$ the next step of the recursion would be ill-defined. □

The classical Bellman equation has $\mathbb{E}[v_{t+1}(x_{t+1}) \mid x_t, u_t]$. Equation (16) replaces this with $\sigma_t(v_{t+1}, x_t, Q_t(x_t, u_t))$. The algorithm structure is identical. Only σ changes.

4 Illustrative Examples

4.1 Example 1: IoT Age-of-Information MDP

4.1.1 Problem Statement

An IoT device observes an autoregressive process $X_{t+1} = X_t + W_t$ where the noise W_t is i.i.d. with distribution

$$\mathbb{P}(W_t = w) = \begin{cases} \frac{1}{5} - \frac{|w|}{25} & |w| \leq 5 \\ 0 & \text{otherwise} \end{cases} \tag{18}$$

This is a triangular distribution on $\{-5, \ldots, 5\}$, symmetric around zero, with $\mathbb{E}[W_t] = 0$. The process starts at $X_1 = 0$.

At each period the device chooses whether to transmit its current observation ($A_t = 1$, cost λ) or remain silent ($A_t = 0$, no cost). When it does not transmit, the receiver holds the last transmitted value Z_t and incurs the staleness penalty $(X_t - Z_t)^2$.

State space. The state $S_t = X_t - Z_t$ is the current staleness error, with dynamics:

$$S_{t+1} = \begin{cases} S_t + W_t & \text{if } A_t = 0 \quad \text{(error accumulates)} \\ W_t & \text{if } A_t = 1 \quad \text{(error resets to fresh noise)} \end{cases} \tag{19}$$

The per-step cost is $c(s, a) = \lambda a + (1 - a)s^2$.

State space truncation. The natural state space is unbounded. We truncate to $S_t \in \{-B, \ldots, B\}$ via $[x]_{-B}^{B} := \max(-B, \min(B, x))$, giving $N = 2B + 1$ states. We use $T = 20$, $\lambda = 100$, $B = 100$. Since the noise support is $[-5, 5]$ and the horizon is $T = 20$, the error grows by at most $5 \times 20 = 100$ from $S_1 = 0$, so $B = 100$ contains the reachable state space exactly.

For the three-method comparison we use $B = 50$ to keep CVaR computationally tractable. The boundary sensitivity analysis in Figure 3 confirms that results are stable for $B \geq 70$.

4.1.2 Risk-Neutral Solution

The risk-neutral finite-horizon Bellman equation is:

$$v_t(s) = \min_{a \in \{0,1\}} \left\{ c(s, a) + \mathbb{E}[v_{t+1}(S_{t+1}) \mid S_t = s, A_t = a] \right\} \tag{20}$$

with $v_{T+1}(s) = 0$. The expectations are matrix-vector products. If $\mathbf{P}^0$ and $\mathbf{P}^1$ are the $N \times N$ transition matrices for actions 0 and 1:

$$\mathbf{q}_t^0 = \mathbf{c}^0 + \mathbf{P}^0 \mathbf{v}_{t+1}, \qquad \mathbf{q}_t^1 = \mathbf{c}^1 + \mathbf{P}^1 \mathbf{v}_{t+1}, \qquad \mathbf{v}_t = \min(\mathbf{q}_t^0, \mathbf{q}_t^1). \tag{21}$$

Threshold structure. The cost $c(s, 0) = s^2$ is symmetric and convex. The noise distribution is symmetric around zero. By symmetry and convexity of the value function, the optimal policy has a time-varying threshold $s_t^* \geq 0$:

$$\hat{\pi}_t(s) = \begin{cases} 1 & |s| \geq s_t^* \\ 0 & |s| < s_t^* \end{cases} \tag{22}$$

The agent transmits when the staleness error is large enough that the cost $\lambda = 100$ is justified by the reduction in future squared-error costs.

4.1.3 Risk-Averse Solution

We compare three risk measures: expected value (baseline), CVaR at $\alpha = 0.3$, and MSD at $\kappa = 1.0$.

CVaR risk transition mapping. CVaR admits both a primal form (minimisation over a scalar threshold θ) and a dual form as a supremum over distorted transition probabilities. We implement the dual directly. For each state s_i the adversary solves:

$$\sigma_{\text{CVaR}}(v, s_i, \mathbf{p}_i) = \max_{\mu \in \mathcal{A}(\mathbf{p}_i)} \sum_j \mu_j v_j, \tag{23}$$

where the adversarial set is:

$$\mathcal{A}(\mathbf{p}_i) = \left\{ \mu : 0 \leq \mu_j \leq \frac{p_{ij}}{\alpha}, \sum_j \mu_j = 1 \right\}. \tag{24}$$

This is Definition 5 and Example 5 of Ruszczyński [2010] made explicit: the adversary concentrates mass on the highest-cost next states, subject to the distortion bound p_{ij}/α per state. The solution is found by a greedy algorithm: sort next-state values v_j in descending order and assign $\mu_j = \min(p_{ij}/\alpha, \text{remaining mass})$ until the total reaches one. No LP solver is needed.

MSD risk transition mapping.

$$\sigma_{\text{MSD}}(v, s_i, \mathbf{p}_i) = \mathbf{p}_i^\top v + \kappa\, \mathbf{p}_i^\top (v - \mathbf{p}_i^\top v)_+. \tag{25}$$

MSD penalises next-state values exceeding the conditional mean. At $\kappa = 1$ the penalty equals the full expected positive deviation.

Complexity. EV and MSD run in $O(T \cdot N^2)$ time. CVaR requires one sort of length N per state per step, giving $O(T \cdot N^2 \log N)$. Because the noise support is bounded to $[-5, 5]$, each row of $\mathbf{P}^0$ and $\mathbf{P}^1$ has at most 11 nonzero entries regardless of N, so the greedy inner loop terminates in at most 11 iterations. In practice CVaR adds negligible overhead over EV and MSD. The three-method comparison uses $B = 50$; boundary sensitivity uses $B \in \{50, 60, 70, 80, 100\}$.

4.1.4 Results

Value functions (Figure 1, top). Both value functions are symmetric and convex in s, consistent with the quadratic no-transmit cost and symmetric noise. The CVaR value function is uniformly steeper: the adversary inflates the probability of noise realisations that push the error from zero, raising the perceived cost at large-error states. As time remaining increases, values increase because more periods of staleness cost accumulate.

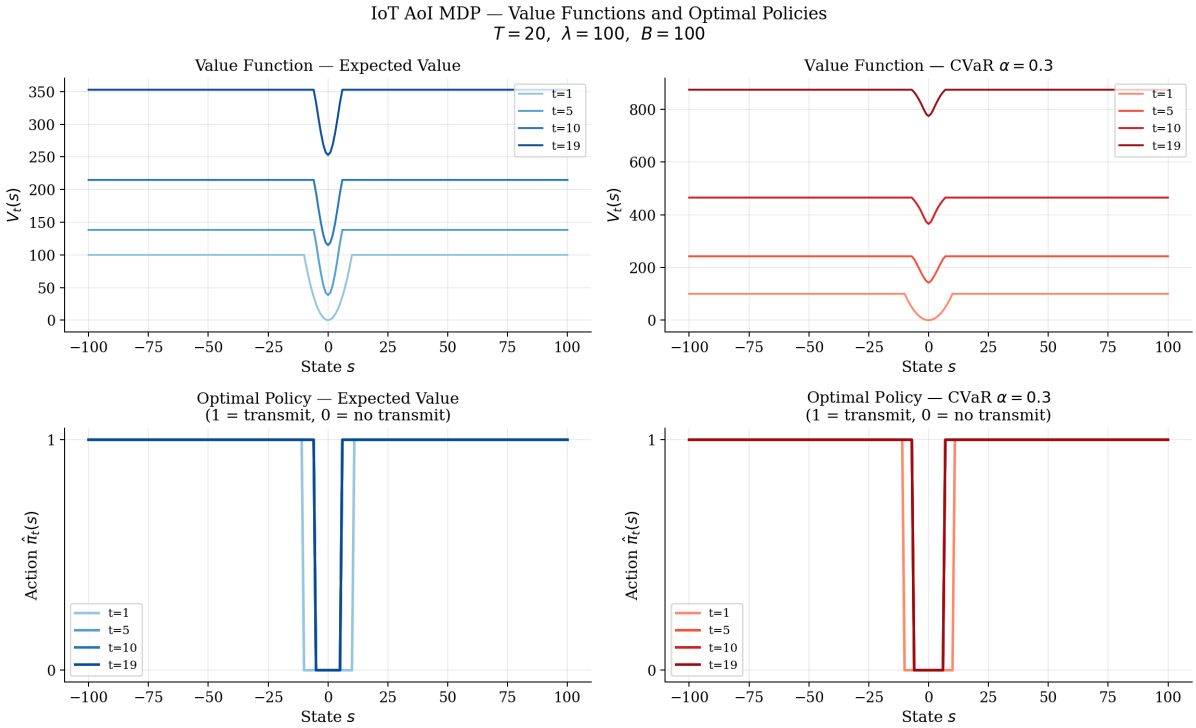


Figure 1: Value functions (top) and optimal policies (bottom) for expected value (left, blue) and CVaR $\alpha = 0.3$ (right, red), at $t \in \{1, 5, 10, 19\}$ steps remaining. $T = 20$, $\lambda = 100$, $B = 50$.

Optimal policies (Figure 1, bottom). Both policies confirm the threshold structure. The CVaR threshold is lower at every time point. Near the deadline ($t = 1$), even moderate errors are not tolerated. With many periods remaining ($t = 19$), the agent can wait longer before transmitting.

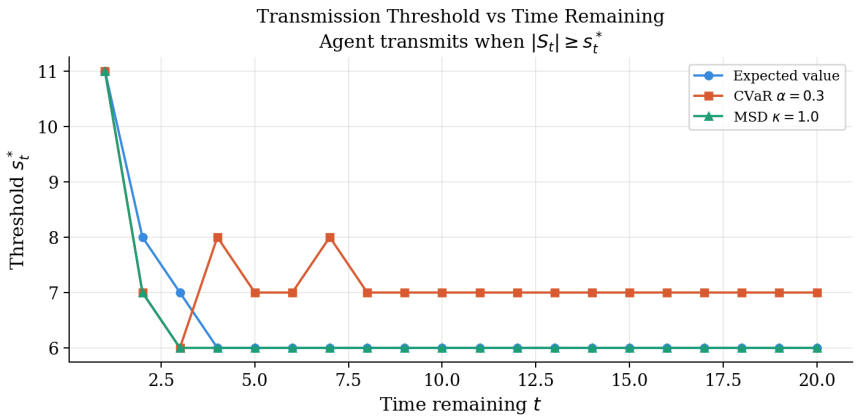


Figure 2: Transmission threshold s_t^* vs. time remaining t for all three risk measures. CVaR and MSD lie uniformly below the EV threshold. Risk-averse agents transmit more eagerly at every stage.

Transmission threshold (Figure 2). All thresholds increase monotonically with time remaining. CVaR uses the lowest threshold throughout, because adversarial inflation of bad noise realisations raises the perceived cost of inaction. MSD falls between EV and CVaR, consistent with its weaker adversarial pressure via a norm-ball rather than polytope risk envelope.

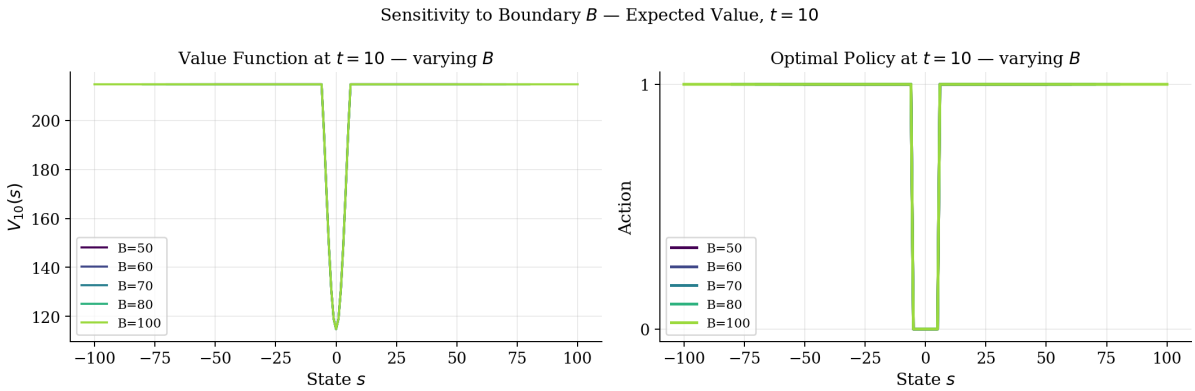


Figure 3: Sensitivity to the truncation boundary $B \in \{50, 60, 70, 80, 100\}$ at $t = 10$, expected value. Value functions (left) and policies (right) converge as B increases. Curves for $B \geq 70$ are nearly indistinguishable, confirming that $B = 100$ introduces negligible error.

Boundary sensitivity (Figure 3). The truncation at B distorts costs and transition probabilities near the boundary. Both value function and policy converge as B grows. With $|W_t| \leq 5$ and $t = 10$ steps remaining, the error reaches at most 50, so any $B \geq 50$ yields the correct policy in the reachable region. Remaining differences in the value function arise from boundary effects propagating backward from earlier steps.

4.1.5 Connection to the Framework

The IoT example illustrates all three theoretical contributions of Ruszczyński [2010].

Time consistency. CVaR and MSD policies are derived from one-step risk measures composed via the nested factorisation (10). The agent never regrets a past transmission decision. This is not guaranteed by naive dynamic application of a static risk measure.

Markov property. Despite the richness of CVaR, the optimal policy depends only on the current state S_t. The risk transition mapping σ restricts the adversary to depend on $(S_t, \mathbf{P}^a_{S_t})$ alone.

Existence of an optimal Markov policy. Conditions (i)–(v) of theorem 2 are satisfied trivially: the state space is finite, the action space $\{0, 1\}$ is compact, costs are bounded, and the transition kernel is continuous. Backward induction terminates in $T = 20$ steps.

4.2 Example 2: Risk-Averse Concert Ticket Pricing

4.2.1 Problem Statement

A firm sells $C = 5$ perishable tickets over $T = 10$ periods. At each period it sets price $p \in \{0, 5, 10, 15, 20, 25\}$. A customer buys with probability $d(p) = \max(0, 1 - p/25)$. Unsold tickets at period T are worthless.

The state is current inventory $c \in \{0, 1, \ldots, 5\}$. Applying Theorem 2 of Ruszczyński [2010] in the maximisation form gives:

$$V_t(c) = \max_p \left\{ d(p) \cdot p + \sigma\left(\begin{bmatrix} V_{t+1}(c) \\ V_{t+1}(c-1) \end{bmatrix}, \begin{bmatrix} 1 - d(p) \\ d(p) \end{bmatrix} \right) \right\}, \quad t = T, \ldots, 1. \tag{26}$$

4.2.2 The Core Tension

Prices $p = 10$ and $p = 15$ both yield expected single-period revenue $p \cdot d(p) = 6$. Price 15 earns nothing 60% of the time; price 10 earns nothing only 40% of the time. A risk-neutral agent is indifferent. A CVaR agent at $\alpha = 0.3$ inflates the no-sale probability at price 15 by a factor of

$1/\alpha = 3.3$, making the low-revenue tail appear far more likely. Price 10 is therefore strongly preferred.

4.2.3 Value Functions

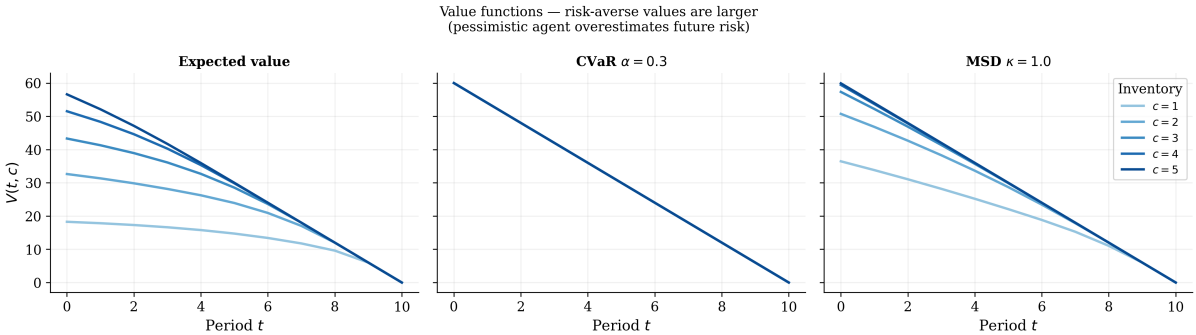
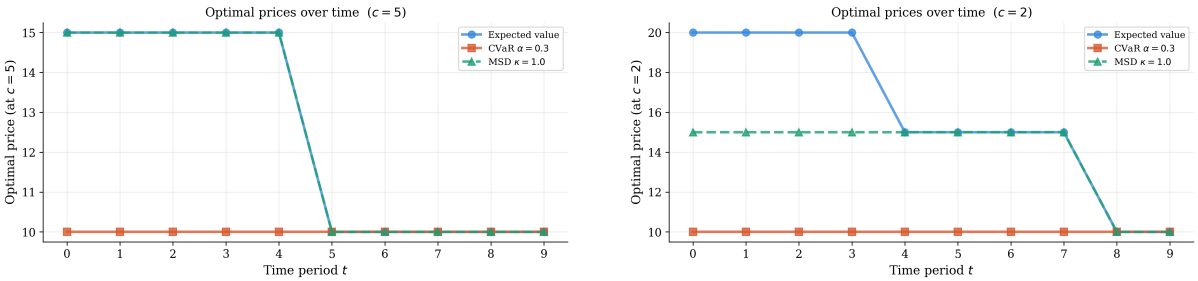


Figure 4: Value functions $V(t, c)$ for all inventory levels $c \in \{1, \ldots, 5\}$ under each risk measure. Risk-averse values are uniformly larger than the EV baseline. The CVaR function is nearly linear in t, reflecting the constant price-10 policy. The EV function is concave, reflecting the late-horizon price switch.

Figure 4 shows three panels, one per risk measure. Risk-averse values are everywhere larger than the EV baseline: the pessimistic agent overweights bad outcomes at every state. The CVaR function is nearly linear in t, a direct consequence of the constant pricing policy.

4.2.4 Optimal Pricing Policies



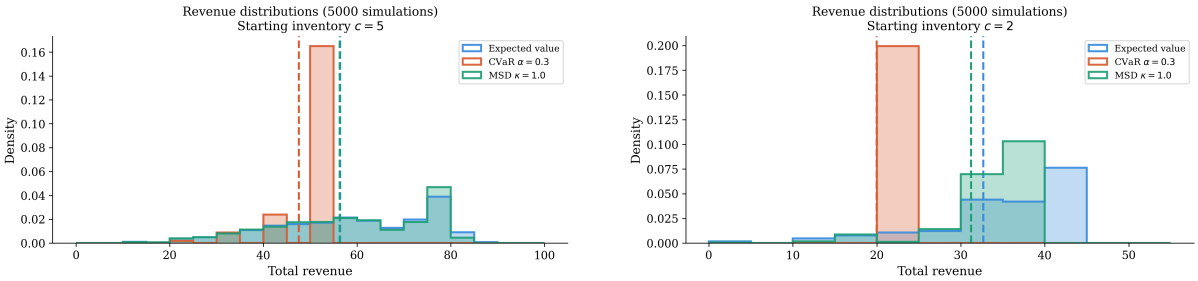
(a) Starting inventory $c = 5$.

(b) Starting inventory $c = 2$.

Figure 5: Optimal price path under each risk measure. CVaR sets price 10 throughout the horizon in both cases. EV uses high prices early and switches late. MSD tracks EV at $c = 5$ but diverges at $c = 2$, where fewer units make tail risk more salient.

At $c = 5$ (Figure 5a), EV and MSD hold price 15 through period 4 and then switch to price 10. CVaR sets price 10 for the full horizon, because the inflated no-sale probability makes the high price unacceptable at every stage.

At $c = 2$ (Figure 5b), differences are more pronounced. EV begins at price 20 and steps down late. MSD holds price 15 through most of the horizon before a late switch. CVaR holds price 10 from the start. With fewer units, each no-sale event is more consequential and the risk envelope punishes high prices more severely. This illustrates a key prediction of Ruszczyński [2010]: risk aversion becomes more behaviourally distinct as the state becomes more constrained.



(a) Starting inventory $c = 5$.

(b) Starting inventory $c = 2$.

Figure 6: Simulated revenue distributions (5000 trajectories). Dashed lines mark the mean. CVaR produces a concentrated, low-variance distribution. At $c = 2$, CVaR eliminates almost all tail variance while EV and MSD retain a wide spread.

4.2.5 Revenue Distributions

Table 1 summarises the quantitative results.

Method	$c = 5$		$c = 2$	
	$\mathbb{E}[\text{Rev}]$	**Semivar**	$\mathbb{E}[\text{Rev}]$	**Semivar**
Expected value	56.4	159.8	32.7	48.0
CVaR $\alpha = 0.3$	47.6	29.4	20.0	0.4
MSD $\kappa = 1.0$	56.3	155.0	31.2	23.0

Table 1: Mean revenue and semivariance from 5000 Monte Carlo simulations. Semivariance: $\mathbb{E}[(\bar{r} - r)_+^2]$.

At $c = 5$, CVaR reduces semivariance by 82% (from 159.8 to 29.4) at the cost of 16% lower mean revenue. MSD preserves the mean almost exactly but barely moves semivariance (155.0 vs. 159.8). MSD's local one-step upside penalty does not compound sufficiently over ten periods to reshape the tail.

At $c = 2$, CVaR reduces semivariance by 99% (from 48.0 to 0.4) by concentrating outcomes near a single value. Price 10 with $d(10) = 0.6$ and two units gives near-certain revenue of 20. EV and MSD retain wide, multi-modal distributions at $c = 2$, corresponding to the variable number of units sold at fluctuating prices.

These results connect directly to the challenge in Section 2. The EV model cannot distinguish two distributions with the same mean but different tails. The risk-averse framework resolves this by building tail-sensitivity into the Bellman recursion via σ, producing policies with verifiably different distributional outcomes.

The Examples are available in a GitHub repository: here

5 Conclusion

5.1 Summary

This report reviewed the framework of Ruszczyński [2010] and Ruszczyński and Shapiro [2006] for risk-sensitive dynamic programming. The two papers together answer a question that the classical theory leaves open: how do you build a recursive, tractable optimisation criterion that is sensitive to tail risk?

Ruszczyński and Shapiro [2006] supply the foundation. Conditional risk mappings satisfying axioms (A1)–(A3) admit a duality representation as worst-case conditional expectations, and their Section 7 shows how to compose them into a multi-period objective with well-posed dynamic programming equations. Ruszczyński [2010] then identifies time consistency as the key structural property: Theorem 1 shows it is equivalent to the nested factorisation, and the risk transition mapping (Definitions 5–6) restricts the conditional risk measure to depend only on the current state and action. Theorem 2 then establishes existence of an optimal Markov policy under mild regularity conditions.

Both examples confirm the theory. The IoT agent transmits more eagerly under CVaR and MSD, with a threshold that is uniformly lower than the EV baseline. The pricing agent under CVaR sets a constant low price, reducing semivariance by 82% at $c = 5$ and 99% at $c = 2$.

5.2 Analysis

Restrictive assumptions. Condition (v) requires compact action sets, which excludes continuous pricing problems in their natural form. Condition (i) requires continuous transition kernels, which excludes systems with jumps or discontinuous dynamics. The average-cost criterion remains open: time consistency lacks a natural terminal anchor in the infinite-horizon setting. The risk envelope $\mathcal{A}$ is assumed known, and learning it from data while preserving time consistency is an open problem.

Computational cost and trade-offs. After switching to the adversarial dual formulation, CVaR requires one sort of length N per backward step rather than an LP, giving $O(T \cdot N^2 \log N)$ overall. Because the noise support is bounded to $[-5, 5]$, each row of $\mathbf{P}^0$ and $\mathbf{P}^1$ has at most 11 nonzero entries, so the greedy inner loop terminates in at most 11 iterations regardless of N. Figure 7 confirms this empirically: all three methods run at comparable wall-clock times, with differences within measurement noise.

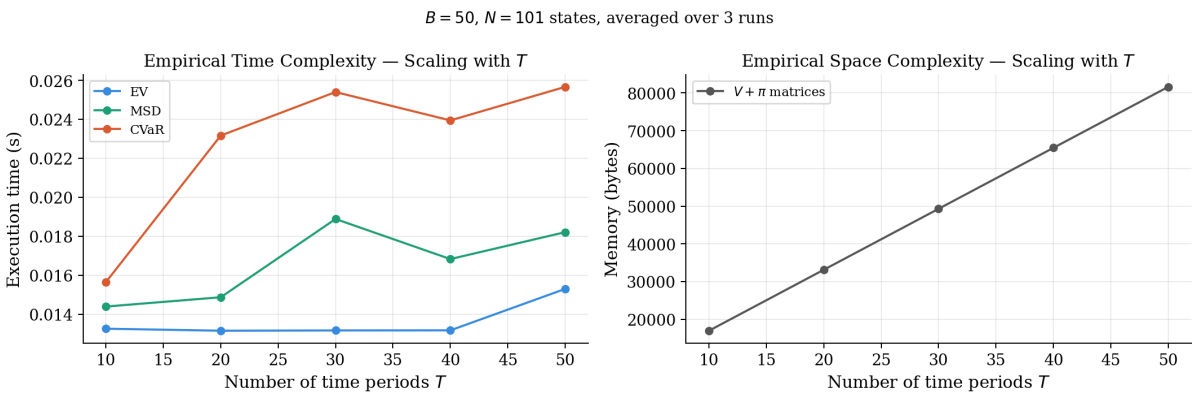


Figure 7: Empirical time and space complexity as T varies, $B = 50$, $N = 101$ states, averaged over 3 runs. All three methods scale linearly in T and run at comparable speeds. The small constant overhead of CVaR over EV reflects the sort operation and does not grow with T. Memory scales linearly as expected from storing $V \in \mathbb{R}^{(T+1)\times N}$ and $\pi \in \mathbb{Z}^{T\times N}$.

Comparison to robust MDPs. The risk-averse DP is structurally equivalent to a minimax robust MDP, as shown in Section 10 of Ruszczyński [2010]. The distinction is that the uncertainty set $\mathcal{A}$ here is derived axiomatically from a coherent risk measure rather than specified by hand. This gives the envelope a behavioural interpretation and guarantees time consistency by construction, whereas a hand-specified robust MDP set carries no such guarantee. The frame-

works are therefore not interchangeable in practice, even when they lead to the same dynamic programming equations.

5.3 Wider Context

The framework has influenced research in several directions. Chow and Ghavamzadeh [2014] extended it to approximate dynamic programming for large state spaces, making the approach practical for problems where exact backward induction is intractable. Tamar et al. [2015] used the nested factorisation to define risk-sensitive return gradients, enabling policy gradient methods for reinforcement learning under CVaR objectives. The connection to Wasserstein distributionally robust MDPs is explored in Shapiro [2021], which shows that certain Wasserstein balls lead to risk measures consistent with the axiomatic framework here. The IoT age-of-information problem under risk-neutral DP is studied in Sun et al. [2017]: the risk-averse extension pursued in Example 1 is, to our knowledge, new and points to a natural direction for future work.

Taken together, the two papers marked a turning point. Before them, risk aversion in MDPs was handled through separable utility functions or by applying static risk measures naively to cumulative costs, both of which either lose time consistency or destroy the Markov structure. The framework of Ruszczyński and Shapiro [2006] and Ruszczyński [2010] showed that these are not necessary concessions: a coherent, time-consistent, Markov-compatible theory of risk-sensitive control is possible and computationally accessible.

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